

Vivekanand B

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SUMMARY

Staff Software Engineer with an intrapreneurial DNA and a multidisciplinary foundation spanning Mechanical, Electronics, and Software Engineering. Expert in applying the precision of aerospace design (ISRO, Safran) to build scalable, enterprise-grade AI products. Currently driving "Zero-to-One" initiatives at Sanas.ai, operating as a technical co-founder within the organization to deliver the Sanas Consumer App and Sales Copilot from concept to GA. A pioneer in AI-Native Development, orchestrating multi-agent workflows to accelerate delivery while bridging the gap between Research, Engineering, and Product strategy.

EXPERIENCE

Staff Engineer · Sanas.ai · Nov 2024 - Present

Speech Intelligence (SI) Platform (Jan 2026 - Present)

- Designed and implemented Speech Intelligence (SI), a privacy-first, edge-powered conversational intelligence and voice analytics platform built directly on top of the company's core Accent Translation (AT) and Noise Cancellation (NC) desktop stack. The system processes high-fidelity audio streams locally to generate real-time operational metrics, automated post-call CRM documentation, and near-real-time compliance alerts without exporting raw audio or customer PII to the cloud.
- High-Throughput Asynchronous Orchestration: Architected a distributed microservice framework backed by a message queue to ingest raw chunks, reconcile channel-wise timestamp overlaps, and store merged data efficiently within Amazon S3.
- Cascading Real-Time Analytics Pipeline: Designed a highly scalable, four-stage analytical data pipeline using a ClickHouse OLAP database. Utilized the AggregatingMergeTree engine paired with state combinators (sumState, argMaxMergeState, uniqState) to incrementally aggregate operational metrics (AHT, agent/customer talk-time ratios, distinct call volumes) across daily, weekly, and monthly rollups without re-scanning raw log data.
- Multi-Track AI Analytics Workers: Developed asynchronous queue-driven workers executing concurrent post-call LLM pipelines. Track A generates concise call summaries, discussion items, and action bullet points. Track B computes automated performance metrics, customer intent hashtags, and best-fit call dispositions.
- Context-Aware Presentation Layer: Built a multi-tier visualization framework consisting of a real-time active call widget, a VDI-whitelisted web portal enabling agents in restricted environments to copy-paste summaries into CRMs, and an administrative dashboard highlighting near-real-time fraud and misrepresentation alerts on a 30-second refresh cycle.
- Expanded conversational data coverage from an industry baseline of 1%-3% to 100% total interaction visibility across all deployed voice operations.
- Delivered structured call artifacts within 4 seconds of termination, systematically driving down After-Call Work (ACW) and overall Average Handle Time (AHT).
- Achieved an alert latency SLA of under 60-120 seconds for flagging high-severity policy breaches and fraud patterns.
- Eliminated multi-gigabyte query out-of-memory (OOM) failures by strict adherence to sorted database primary keys and partition-aware query filtering.

Playground - Unified Audio Transformation Showcase Platform (Dec 2025 - Mar 2026)

- Unified AI Product Suite Integration: Integrated distributed machine learning models to orchestrate noise cancellation, voice isolation, speech enhancement, and language translation simultaneously within a single user session.
- Enforced hard file constraints (max 30-second duration and 10MB file limits) at ingress to tightly regulate production cloud costs and protect against abuse.

Unified ML Platform - Cross-Functional Orchestration (Dec 2025 - Feb 2026)

- Project Overview: Orchestrated the development of a centralized ML Platform to unify disparate workflows across DevOps, MLOps, and Innovation teams. I led a cross-functional team of 5 engineers to replace fragmented tooling with a standardized, scalable infrastructure.
- Unified 3 disparate teams (DevOps, MLOps, Innovation) under a single technical vision, eliminating siloed workflows.

Language Translation

- Cross Platform App – B2C (Sanas Mobile) Aug 2025 – Nov 2025 | Staff Engineer | Sanas.ai
- Project Overview: Led the rapid development of the Sanas Consumer Mobile App, delivering a launch-ready React Native application in under 4 weeks. Achieved 100% iOS/Android parity and global readiness with zero prior mobile experience by pioneering an AI-orchestrated development workflow.

Project 2: Workspace Time Zone Selector – Cloud Portal

- Mar 2025 Full-Stack Engineer | Sanas.ai
- Project Overview: Designed and implemented a workspace-level time zone management system for the Sanas Admin Portal, resolving long-standing timestamp inconsistencies across distributed regions (India, Philippines, U.S.). This feature ensured 100% display consistency, aligning global operations with internal analytics through a standardized UTC-based data framework.
- Audit & Validation: Conducted a full production schema audit to ensure UTC compliance. Authored and executed a multi-region E2E test strategy covering DST edge cases and transitions across Manila, India, and U.S. zones.
- Achieved 100% consistency in timestamp display across all admin regions.
- Reduced user confusion and support tickets related to time mismatches by over 60%.

Project 1: Internal Tools Portal & Application Modernization

- Project Overview: Addressed critical issues of "tool sprawl" and inconsistent documentation by taking ownership of over 25 legacy internal applications and building a new, centralized portal to house them.
- Leveraged GitHub Copilot extensively to rapidly learn and diagnose over 25 disparate codebases written in React, Vue, Python, and Node.js, achieving proficiency in weeks instead of months.
- Designed and implemented a standardized, scalable framework for all future internal tool development, ensuring consistency and maintainability.
- Improved Productivity: Reduced the time for engineers to find tools and documentation from hours to seconds.
- Reduced Onboarding Time: The portal was projected to cut the technical onboarding time for new engineers by 50%.
- Established Governance: Created a single source of truth that eliminated redundant tools and standardized operations.

Project 2: MLOps Data Observability Dashboard

- Project Overview: Developed a comprehensive data observability dashboard to provide the first-ever end-to-end visualization of the company's complex MLOps data pipeline, enabling deep analysis and strategic improvements.

Software Engineer • NovaSignal/NeuraSignal • Nov 2020 – Nov 2024

Jira Analyzer

- Automated analysis of 33% of the total Jira issues (23/69).
- Identified key system performance and navigation/UI issues that helped prioritize fixes (e.g., 12 system performance issues, 3 navigation/workflow issues).

SKILLS

Databases: PostgreSQL

Languages: Python, JavaScript, TypeScript

Technology: IoT

OS: Windows, Linux, Raspberry Pi, MacOS

PDM: PDM Link SSCI, ENNOVIA VPM

Cloud: GCP (VM, Serverless, VPN, Setup), AWS (EC2)

EDUCATION

- Bachelor of Engineering (Mechanical Engineering) | 2007-2011 | M.V.J.College of Engineering (Visvesvaraya Technological University) | First Class with Distinction | 78.26%
- PUC in Science {Physics, Chemistry, Maths, Biology} | St. Joseph's Pre-University College (Department of Pre University Education, Karnataka) | 2005-2007 | 61.83%
- SSLC | St. Joseph's Indian High School (Karnataka Secondary Education Board) | 2005 | 78.88%